

20.1 Kinds of residuals

The model makes its assumptions about the errors, which are never observed. The residuals are their visible stand-ins, imperfect in a precise way: they are the projection of the errors onto a subspace of dimension $n - p$, so they are correlated, have unequal variances, and are shrunk most at the cases that matter most. This section defines rescaled residuals, finds their laws, and asks what residual plots reveal.

20.1.1 The setting

Throughout the chapter,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I},$$

with $\mathbf{X}$ of size $n \times p$ and full column rank $p < n$. Results that need normal errors say so and then assume Eq. 7.3.1. The hat matrix is $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, and h_{ij} denotes its (i, j) entry, so that h_{ii} is the leverage of case i (Definition 6.8.1). The i th row of $\mathbf{X}$, written as a column, is $\mathbf{x}_{(i)}$, and $\mathbf{e}_i$ is the i th coordinate vector of $\mathbb{R}^n$. The residual vector is $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y}$, with entries $\hat{\varepsilon}_i$, and $s^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2 / (n - p)$.

Everything in the chapter depends on $\mathbf{X}$ only through $\mathcal{C}(\mathbf{X})$. If $\mathbf{X}$ has rank $r < p$, replace p by r in every degree of freedom and every formula that involves $\mathbf{M}$ alone. Formulas that involve $(\mathbf{X}^\top \mathbf{X})^{-1}$ hold for any full-rank reparameterization (Theorem 6.7.1).

Two facts from earlier chapters drive everything that follows. First, by Proposition 6.4.3(e),

$$\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}, \tag{20.1.1}$$

so the residuals are a fixed linear image of the errors. Second, by Proposition 5.5.3(b), $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$. So $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$ and, for $i \ne$

(c) the **internally studentized residual** is

$$r_i = \frac{\hat{\varepsilon}_i}{s\sqrt{1 - h_{ii}}};$$

(d) the **externally studentized residual** is

$$t_i = \frac{\hat{\varepsilon}_i}{s_{(i)}\sqrt{1 - h_{ii}}},$$

where $s_{(i)}^2 = \text{SSE}_{(i)}/(n - p - 1)$ is the residual mean square of the least squares fit to the $n - 1$ cases other than i .

Under normal errors z_i is exactly $N(0, 1)$, but it cannot be computed. The studentized residuals replace σ by an estimate: s , which includes case i , or $s_{(i)}$, which does not, so that a gross error in y_i cannot inflate the yardstick it is measured against. [Section 20.3](#) shows that $s_{(i)}$ costs nothing extra and that t_i has an exact t law.

Remark (Names). Terminology varies. Some authors call r_i , or even $\hat{\varepsilon}_i/s$, the standardized residual, and t_i is also called the studentized deleted residual or RSTUDENT. R's `rstandard` and `rstudent` return r_i and t_i , as do `resid.studentized.internal` and `resid.studentized.external` in `statsmodels`.

20.1.3 The law of the internally studentized residual

The residual r_i is not t -distributed, because $\hat{\varepsilon}_i$ is part of s . Its law is a rescaled beta.

Proposition 20.1.1 — Law of the internally studentized residual

Assume Eq. 7.3.1, $n - p \geq 2$ and $h_{ii} < 1$. Then:

- (a) $r_i^2/(n - p) \sim \text{Beta}(\frac{1}{2}, \frac{n-p-1}{2})$;
- (b) $|r_i| \leq \sqrt{n - p}$, $E(r_i) = 0$ and $\text{Var}(r_i) = 1$;
- (c) the law of r_i is the same for every case and does not depend on $\mathbf$

together with the symmetry, determines the law of r_i .

■

The bound in (b) bites when $n - p$ is small: with $n - p = 5$, no $|r_i|$ can exceed $\sqrt{5} \approx 2.236$, however wild the case, because a gross error inflates s along with $\hat{\varepsilon}_i$. The r_i are not independent either; their correlations are those of Eq. 20.1.2 (Exercise (C1)).

20.1.4 What a residual plot can show

A residual plot shows residuals against the fitted values, a regressor, an omitted variable or the order of collection, to reveal structure the model has missed. Least squares removes some structure automatically, and it is worth knowing exactly which.

Proposition 20.1.2 — What least squares leaves in the residuals

Suppose $\mathbf{1} \in \mathcal{C}(\mathbf{X})$.

- (a) The residual vector is orthogonal to $\mathbf{1}$, to every column of $\mathbf{X}$ and to $\hat{\mathbf{Y}}$. Hence the least squares line through a plot of the residuals against the fitted values, or against any regressor in the model, is the horizontal axis.
- (b) The least squares line through a plot of the residuals against the responses y_i has slope $1 - R^2$.
- (c) If in fact $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$ for some $\boldsymbol{\delta} \in \mathbb{R}^n$, and $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$, then $E(\hat{\boldsymbol{\varepsilon}}) = (\mathbf{I} - \mathbf{M})\boldsymbol{\delta}$ and $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$.

Proof.

- (a) is Proposition 6.4.3(a) and (b). A line fitted with an intercept to points $(a_i, \hat{\varepsilon}_i)$ has slope $\sum_i (a_i - \bar{a})\hat{\varepsilon}_i / \sum_i (a_i - \bar{a})^2$, and the numerator is $\mathbf{a}^\top \hat{\boldsymbol{\varepsilon}} - \bar{a} \mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$ when $\mathbf{a}$ is $\hat{\mathbf{Y}}$ or a column of $\mathbf{X}$.
- (b) With $\mathbf{a} = \mathbf{y}$ the numerator is $\mathbf{y}^\top \hat{\boldsymbol{\varepsilon}} = (\hat{\mathbf{Y$

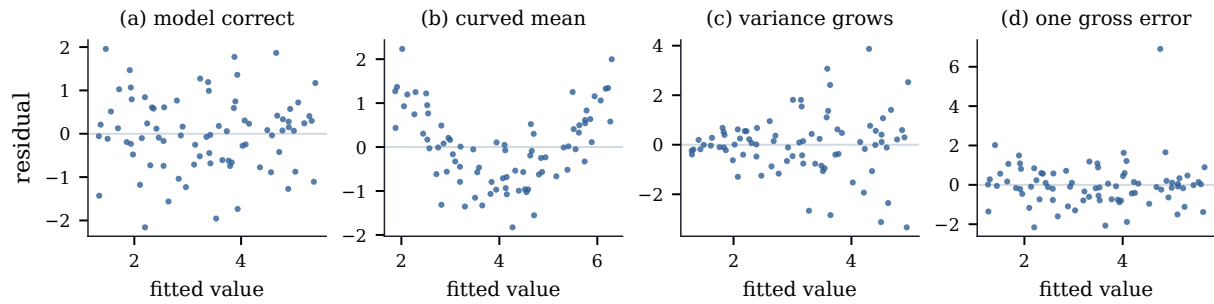


Figure 20.1.1: Residuals against fitted values for four straight-line fits: (a) correct model, (b) curved mean, (c) variance growing with the mean, (d) one gross error.

The standard plots are these. Against **fitted values**: curvature suggests a misspecified mean, a fan shape variance that grows with the mean, isolated points outliers. Against **each regressor**: curvature points to that regressor. Against a variable **not in the model**: better replaced by the added-variable plot of [Section 6.6](#), whose slope is the coefficient the variable would receive ([Exercise \(B1\)](#)). Against **collection order**: runs of equal sign suggest serial correlation, tested in [Chapter 21](#) (Theorem 21.5.3). A **normal quantile plot**: curvature suggests skewed or heavy-tailed errors (Proposition 21.1.1). Studentized residuals are preferable to raw ones throughout, since raw residuals make high-leverage cases look better than they are.

Example. (Four residual plots)

Each panel of [Figure 20.1.1](#) comes from 80 simulated cases with x uniform on $(0, 10)$, fitted by a straight line. In (a) the model is right and the residuals form a structureless band. In (b) the mean has a quadratic term, and the residuals show the parabola the line could not absorb. In (c) the error standard deviation grows with x , and the band opens into a fan. In (d) one response has an error of 8σ ; it stands out, and the line has been pulled slightly towards it. In every panel the residuals are orthogonal to the fitted values, as [Proposition 20.1.2\(a\)](#) requires; the listing prints the inner product.

```
import numpy as np

def ls_fit(X, y):
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    return beta, X @ beta, y - X @ beta

rng = np.random.default_rng(7)
n = 80
x = rng.uniform(0, 10, n)
eps = rng.normal(size=n)
X = np.column_stack([np.ones(n), x])
responses = {
    "(a) model correct": 1 + 0.5 * x + eps,
    "(b) curved mean": 1 + 0.5 * x + 0.08 * (x - 5) ** 2 + 0.6 * eps,
    "(c) variance grows": 1 + 0.5 * x + (0.1 + 0.25 * x) * eps,
    "(d) one gross error": 1 + 0.5 * x + eps + 8.0 * (np.arange(n) == 17),
}
for name, y in responses.items():
    beta, fitted, resid = ls_fit(X, y)
    print(f"{name:22s} slope {beta[1]:.
```

[Run this code online](#)

20.1.5 Partial residuals

A plot of residuals against a regressor $\mathbf{x}_j$ in the model has a flat least squares line ([Proposition 20.1.2\(a\)](#)). The **component-plus-residual plot**, or partial residual plot, adds the fitted linear component back. It plots

$$\hat{\varepsilon}_i + \hat{\beta}_j x_{ij} \quad \text{against} \quad x_{ij}, \quad i = 1, \dots, n.$$

Compare it with the added-variable plot of [Section 6.6](#), which plots $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{y}$ against $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$, where $\mathbf{M}_{(j)}$ projects onto the span of the columns other than $\mathbf{x}_j$.

Proposition 20.1.3 — Two plots with the same line

Let $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, and let $\mathbf{x}_j$ be a column of $\mathbf{X}$ that is not a multiple of $\mathbf{1}$.

- (a) The least squares line of $\hat{\varepsilon} + \hat{\beta}_j \mathbf{x}_j$ on $[\mathbf{1}, \mathbf{x}_j]$ has intercept 0, slope $\hat{\beta}_j$, and residual vector $\hat{\varepsilon}$.
- (b) The least squares line through the origin of $\tilde{\mathbf{y}}$ on $\tilde{\mathbf{x}}_j$ has slope $\hat{\beta}_j$ and residual vector $\hat{\varepsilon}$.

Proof.

- (a) The vector $\hat{\varepsilon}$ is orthogonal to $\mathbf{1}$ and to $\mathbf{x}_j$ by [Proposition 20.1.2\(a\)](#). So $\hat{\varepsilon} + \hat{\beta}_j \mathbf{x}_j = \hat{\beta}_j \mathbf{x}_j + \hat{\varepsilon}$ is the sum of a vector in $\mathcal{C}([\mathbf{1}, \mathbf{x}_j])$ and a vector orthogonal to that space. By Theorem 6.4.1 the first is the fitted vector and the second the residual.
- (b) is Theorem 6.6.2(a) and (b) with $\mathbf{X}_1 = \mathbf{x}_j$.

■

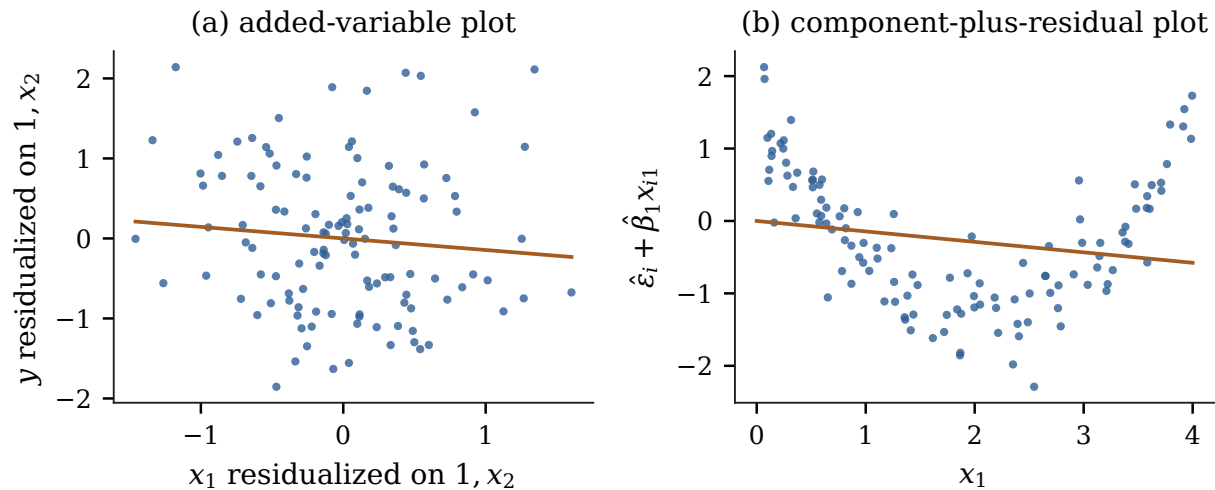


Figure 20.1.2: A curved effect of x_1 , fitted linearly: (a) added-variable plot, (b) component-plus-residual plot. Same slope, same residuals (Proposition 20.1.3).

```

x1 = rng.uniform(0, 4, n2)
x2 = 0.8 * x1 + rng.normal(0, 0.6, n2)           # correlated with x1
y2 = 1 + x2 + 0.7 * (x1 - 2) ** 2 + rng.normal(0, 0.4, n2)
X2 = np.column_stack([np.ones(n2), x1, x2])      # the fitted model is linear in
x1
beta2, fitted2, resid2 = ls_fit(X2, y2)

component = resid2 + beta2[1] * x1                # component-plus-residual for x1
Z = np.column_stack([np.ones(n2), x2])          # everything except x1
x1_tilde = x1 - Z @ np.linalg.lstsq(Z, x1, rcond=None)[0]
y_tilde = y2 - Z @ np.linalg.lstsq(Z, y2, rcond=None)[0]  # added-variable
coordinates
print("coefficient of x1:", beta2[1])
print("slope in component-plus-residual plot:", ls_fit(np.column_stack([np.ones(n2),
x1]), component)[0][1])
print("slope in added-variable plot:", (x1_tilde @ y_tilde) / (x1_tilde @ x1_tilde))

```

[Run this code online](#)

20.1.6 Exercises

20.1.6.1 A. Check your understanding

Exercise A1.

A straight line is fitted to $n = 3$ points. Show that every internally studentized residual satisfies $|r_i| = 1$ (unless all three residuals are zero), whatever the data. What does this say about using r_i to judge a fit with one residual degree of freedom?

Solution. Here $n - p = 1$, so $\mathcal{C}(\mathbf{X})^\perp$ is spanned by one unit vector $\mathbf{v}$, and $\hat{\mathbf{e}} = c\mathbf{v}$ for a scalar c . Then $\hat{e}_i = cv_i$, $s^2 = c^2$ and $1 - h_{ii} = v_i^2$ (the i th diagonal entry of $\mathbf{I} - \mathbf{M} = \mathbf{v}\mathbf{v}^\top$). So $r_i^2 = c^2 v$

Exercise A2.

Show that $\sum_i (1 - h_{ii}) r_i^2 = n - p$ for any data. Deduce that if every leverage is at most $h_{\max} < 1$, then at most $(n - p)/((1 - h_{\max})c^2)$ cases can have $|r_i| > c$.

20.1.6.2 B. Practice**Exercise B1.**

Let $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, let $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$ be a candidate regressor, let $\hat{\gamma}$ be its coefficient when it is added to the model, and let R_z^2 be the R^2 of the regression of $\mathbf{z}$ on $\mathbf{X}$. Show that the least squares slope of the plot of $\hat{\mathbf{e}}$ against $\mathbf{z}$ is $\hat{\gamma}(1 - R_z^2)$. Why is the added-variable plot the better tool for judging whether $\mathbf{z}$ belongs in the model?

Solution. The slope is $\sum_i (z_i - \bar{z})\hat{e}_i / S_{zz} = \mathbf{z}^\top \hat{\mathbf{e}} / S_{zz}$, since $\mathbf{1}^\top \hat{\mathbf{e}} = 0$. Put $\tilde{\mathbf{z}} = (\mathbf{I} - \mathbf{M})\mathbf{z}$. Then $\mathbf{z}^\top \hat{\mathbf{e}} = \mathbf{z}^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \tilde{\mathbf{z}}^\top \mathbf{y}$, and by Theorem 6.6.2(a), $\hat{\gamma} = \tilde{\mathbf{z}}^\top \mathbf{y} / \|\tilde{\mathbf{z}}\|^2$. So $\mathbf{z}^\top \hat{\mathbf{e}} = \hat{\gamma} \|\tilde{\mathbf{z}}\|^2$, and $\|\tilde{\mathbf{z}}\|^2 = S_{zz}(1 - R_z^2)$ is the residual sum of squares of $\mathbf{z}$ on $\mathbf{X}$ (Exercise (B1)). The slope is $\hat{\gamma}(1 - R_z^2)$. When $\mathbf{z}$ is highly correlated with the columns of $\mathbf{X}$, the plot against $\mathbf{z}$ shrinks the effect towards zero, while the added-variable plot shows $\hat{\gamma}$ itself, against the horizontal spread $\|\tilde{\mathbf{z}}\|$ that actually carries information about it.

Exercise B2.

Simple linear regression is fitted to $x_i = i$, $i = 1, \dots, n$, but $E(Y_i) = \beta_0 + \beta_1 x_i + \gamma x_i^2$. Use Proposition 20.1.2(c) to show that the expected residuals lie on a parabola in i , and find its vertex.